

Dirichlet-Multinomial Error-Choice Simulation

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1 Overview

This report evaluates **dirichlet-multinomial** sampling error across cell-type proportions generated from a weighted Beta(, 1) distribution.

1.1 Success rule

For each replicate and cell type j , the estimated proportions are averaged over the N people in that replicate. The average is compared with the population-level true proportion p_j :

$$\bar{p}_j = \frac{1}{N} \sum_{i=1}^N \hat{p}_{ij}, \quad AE_j = |\bar{p}_j - p_j|, \quad ARE_j = \frac{|\bar{p}_j - p_j|}{p_j}. \quad (1)$$

A replicate **succeeds** for AE if $\max AE_j \leq \tau_{AE}$, and for ARE if $\max ARE_j \leq \tau_{ARE}$. AE and ARE are always evaluated separately and never combined into one criterion.

2 Simulation setup

The data are generated in two steps (a Dirichlet-multinomial hierarchy):

1. Person i has a true composition $\theta_i \sim \text{Dirichlet}(c \cdot p)$, where p is the population mean composition and c is the **concentration**.
2. From that person, n cells are sampled: $\text{counts}_i \sim \text{Multinomial}(n, \theta_i)$, so $\hat{p}_{ij} = \text{count}_{ij}/n$.

The population proportions p are fixed values derived from a Beta distribution (Table 2). A larger α makes the composition more skewed, so the rarest types become rarer. Each scenario (α, N, n) is simulated B times. Table 1 lists all settings.

Table 1: Simulation settings (`simulation_dm_errorchoice_defaults()`).

Parameter	Value
alpha (Beta shape)	2, 3, 4, 5
K (cell types)	10
B (replicates per scenario)	1,000
Concentration c	10000
N (people per replicate)	1, 2, 3, 5, 10
n per person, sweep	100 to 100,000,000 (25 log-spaced values)
n per person, threshold sweep	200,000
tau_AE (fixed)	0.005
tau_ARE (fixed)	0.5
Target success rate	0.95
Seed	260926

Table 2: Population mean proportions p_j per cell type (columns) for each α (rows).

alpha	1	2	3	4	5	6	7	8	9	10
2	0.01	0.03	0.05	0.07	0.09	0.11	0.13	0.15	0.17	0.19
3	0.00075	0.0068	0.019	0.037	0.061	0.091	0.13	0.17	0.22	0.27
4	5e-05	0.0014	0.0063	0.017	0.037	0.067	0.11	0.17	0.25	0.34
5	3.2e-06	0.00026	0.002	0.0076	0.021	0.046	0.09	0.16	0.26	0.41

2.1 Concentration c

The **concentration** c controls the **biological variation** between people. The individual proportion θ_{ij} comes from a $Dirichlet(N, c \cdot p_j)$ such that:

$$E(\theta_{ij}) = p_j, \quad \text{Var}(\theta_{ij}) = \frac{p_j(1-p_j)}{c+1}. \quad (2)$$

So c acts as a precision parameter:

- **Small c :** people differ a lot. $\text{Var}(\theta_{ij})$ comes close to its maximum $p_j(1-p_j)$.
- **Large c :** everybody is close to p . As $c \rightarrow \infty$, the model reduces to a plain multinomial with no differences between people.

Conversely, if the between-person variance of a cell type is known from real data, $c = p_j(1-p_j) / \text{Var}(\theta_{ij}) - 1$.

Table 3 shows how large this variation is on the proportion scale, through the standard deviation $\text{SD}(\theta_{ij})$ and the coefficient of variation $\text{CV} = \text{SD}(\theta_{ij}) / p_j = \sqrt{(1-p_j)/(p_j(c+1))}$.

Table 3: Between-person variability of the individual proportions: $\text{SD}(\theta_{ij})$ with the coefficient of variation $\text{CV} = \text{SD} / p_j$ in brackets, for several concentrations c (columns) and population proportions p_j (rows).

	$c = 1,000$	$c = 10,000$	$c = 100,000$	$c = 1,000,000$
$p = 0.01$	0.0031 (0.31)	0.00099 (0.099)	0.00031 (0.031)	0.000099 (0.0099)
$p = 0.05$	0.0069 (0.14)	0.0022 (0.044)	0.00069 (0.014)	0.00022 (0.0044)
$p = 0.1$	0.0095 (0.095)	0.003 (0.03)	0.00095 (0.0095)	0.0003 (0.003)
$p = 0.4$	0.015 (0.039)	0.0049 (0.012)	0.0015 (0.0039)	0.00049 (0.0012)

At the simulated concentration $c = 10,000$, a cell type with a population mean of 10% has a between-person SD of 0.003, so 95% of people have between 9.4% and 10.6% of that type. For rare types the

relative spread is larger. For $p_j = 0.01$ (CV 0.099), 95% of people have between 0.81% and 1.20%. Only for the very rarest types, where $c p_j \ll 1$ (down to $c p_j \approx 0.032$ at $\alpha = 5$), is the distribution of θ_{ij} heavily skewed. There, almost everyone has essentially none of that cell type, and a few people have a lot.

3 Simulation results

3.1 Success rate versus threshold

Figure 1 and Figure 2 show the success rate as a function of the threshold τ at a fixed 200,000 cells per person.

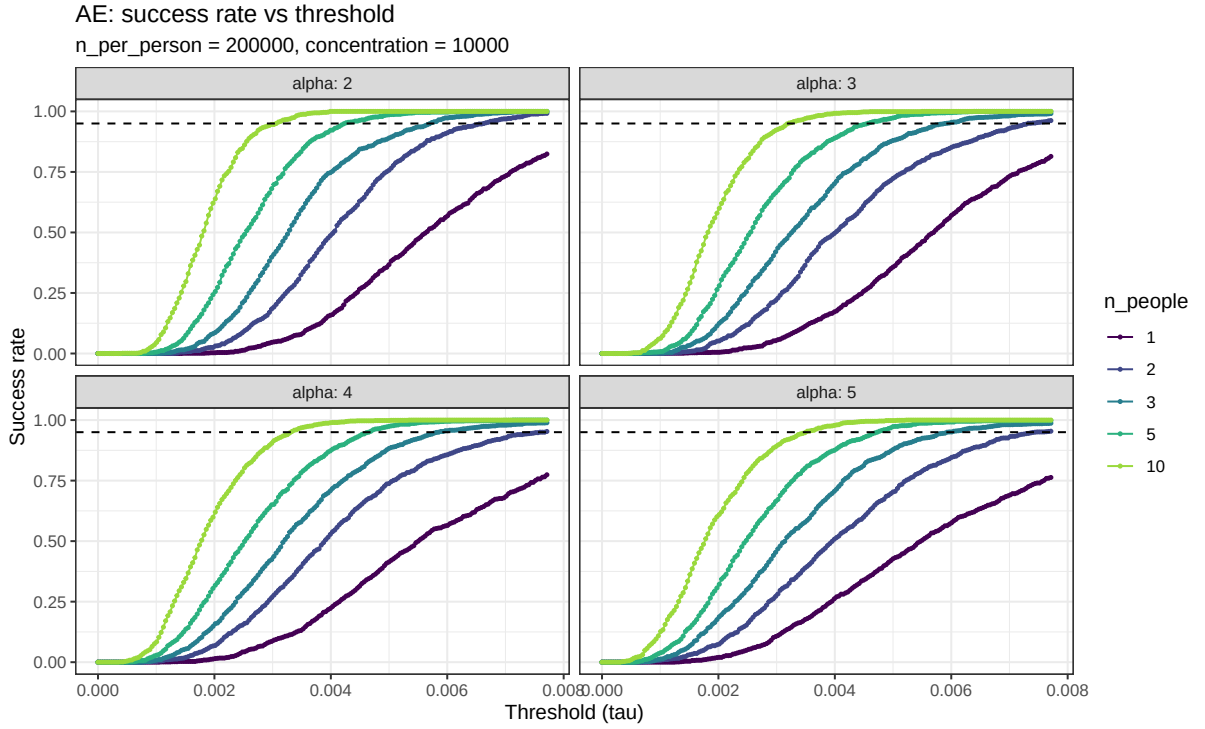


Figure 1: AE success rate versus threshold τ_{AE} at 200,000 cells per person. One panel per α , one colour per number of people N ; the dashed line marks the 95% target.

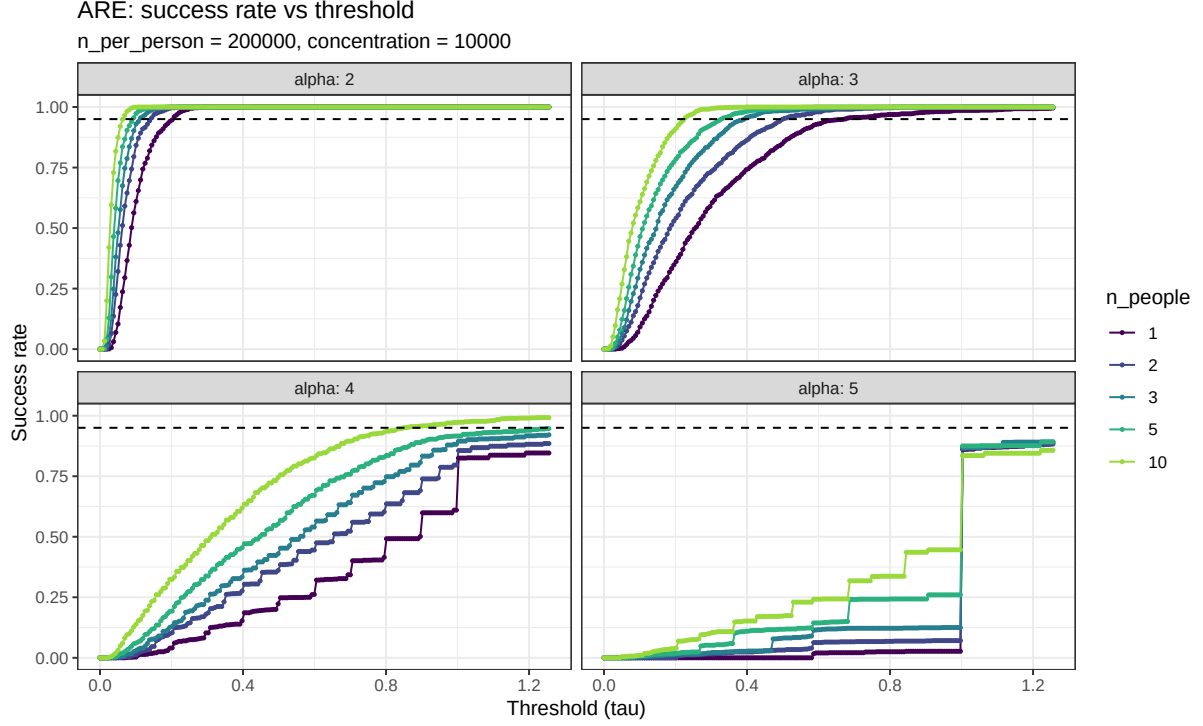


Figure 2: ARE success rate versus threshold τ_{ARE} at 200,000 cells per person. Note the jump at $\tau_{ARE} = 1$.

3.2 Success rate versus cells per person

Figure 3 and Figure 4 fix the threshold ($\tau_{AE} = 0.005$, $\tau_{ARE} = 0.5$) and vary the number of cells per person.

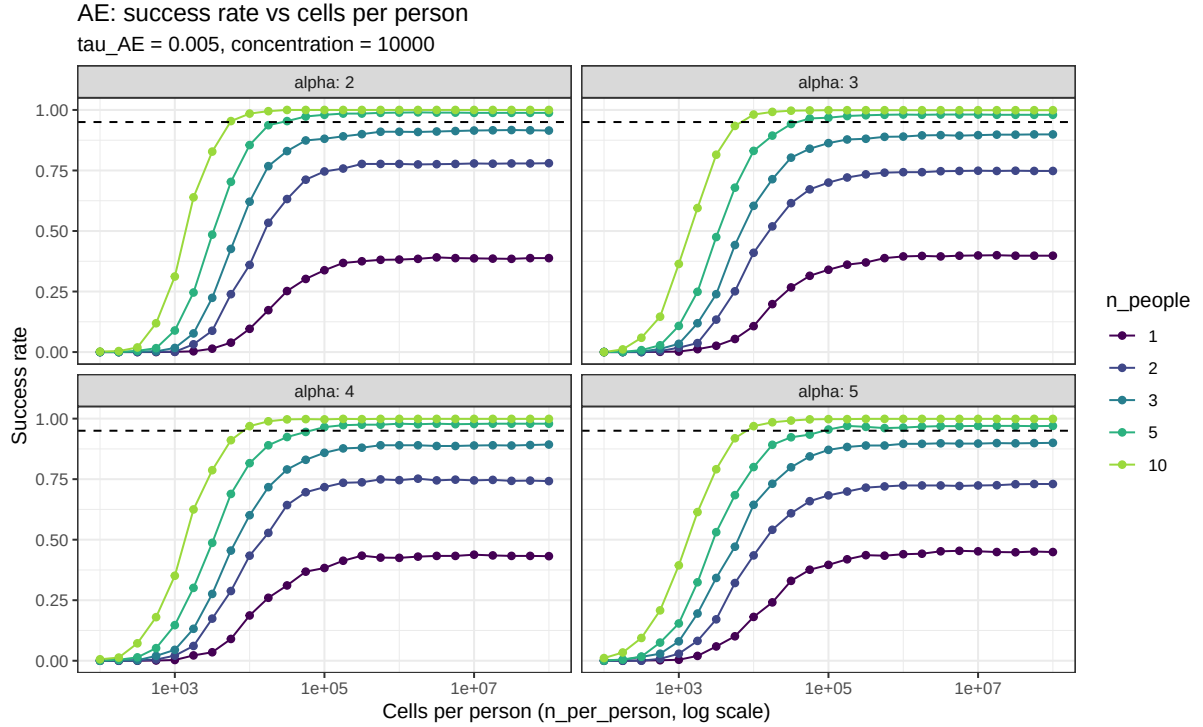


Figure 3: AE success rate versus cells per person (log scale) at $\tau_{AE} = 0.005$. Success rises with cells per person and levels off from roughly $n \approx c$ onwards.

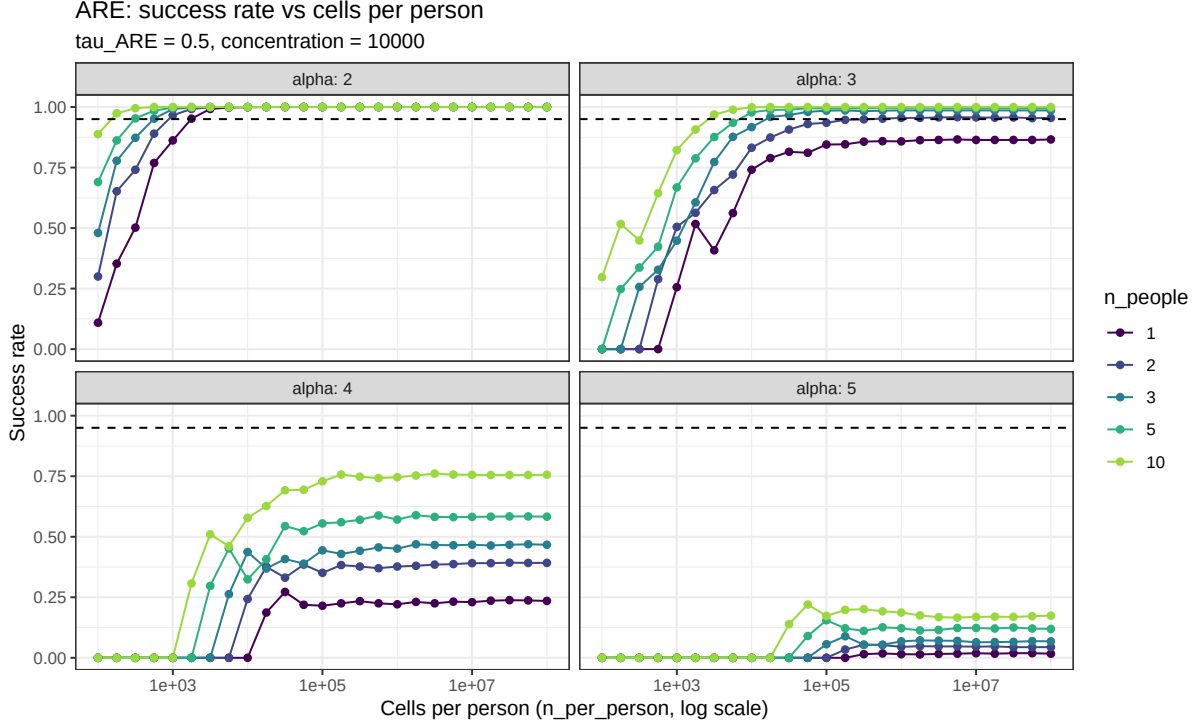


Figure 4: ARE success rate versus cells per person (log scale) at $\tau_{ARE} = 0.5$.

For AE, the success rate increases with the number of cells per person, but only up to a point. At $\alpha = 5$ with a single person, it rises from 0% at 10^2 cells to 18% at $n \approx c$, and ends at 45% at 10^8 cells. With 10 people the same curve ends at 100%. Beyond roughly 10^5 cells per person, extra cells hardly help, and only more people raise the plateau (Section 4).

For ARE the picture depends on α . At $\alpha = 2$ with a single person, success rises from 11% to 100%. At $\alpha = 5$ it ends at only 2% (1 person) and 17% (10 people). The rarest cell types have $c p_j \ll 1$, so in most replicates none of the people carry them. Then $\bar{p}_j = 0$ and $ARE_j = 1 > \tau_{ARE}$, however many cells are counted. At small n the same happens through counting: a rare type that people do carry is often simply not observed. The irregular, non-monotone parts of the curves for $\alpha = 3$ and 4 (around 10^3 to 10^4 cells) come from this discreteness. When the expected count of a rare type per person is of order one, $\hat{p}_{ij}$ can only take a few values, and whether $ARE_j \leq \tau_{ARE}$ jumps with n .

4 Why the curves level off

4.1 Error sources

Given the two-step generation in Section 2, the error of $\bar{p}_j$ has two parts:

- **Differences between people (biological variation).** Each θ_{ij} scatters around p_j with variance $p_j(1 - p_j)/(c + 1)$ (Equation 2).
- **Sampling (technical variation).** Given the person, each $\hat{p}_{ij}$ scatters around θ_{ij} with variance $\theta_{ij}(1 - \theta_{ij})/n$. On average over people, this is $E[\theta_{ij}(1 - \theta_{ij})]/n = \frac{p_j(1 - p_j)c}{(c + 1)n}$.

Assuming people are independent, averaging over N of them gives

$$\text{Var}(\bar{p}_j - p_j) = p_j(1 - p_j) \left[\underbrace{\frac{1}{N(c + 1)}}_{\text{biological}} + \underbrace{\frac{c}{(c + 1)Nn}}_{\text{technical}} \right]. \quad (3)$$

- The **biological term** depends only on the number of people N and the concentration c .
- The **technical term** depends only on the **total** number of cells counted, $N \cdot n$.

The ratio of technical to biological variance is c/n . So technical variation makes up a share $c/(c+n)$ of the variance:

- For $n \ll c$, technical variation dominates, and more cells per person clearly improve the estimate. This is the rising part of Figure 3 and Figure 4.
- At $n = c = 10,000$ the two parts are equal.
- At $n = 10^5$ the technical share is only 9%. From there on, the SD of $\bar{p}_j$ is within 5% of its floor, and the curves level off.
- Even with infinitely many cells per person, $\bar{p}_j$ converges to the average composition of *those particular* N people, and not to p_j . The biological term stays, and only more people lower the plateau.

4.2 Illustration with Additional Simulation

The main simulation only stores the worst-case error per replicate. To illustrate Equation 3 directly per cell type, the report runs a small extra simulation. For $\alpha = 5$ it draws $\theta_{ij} \sim \text{Beta}(c p_j, c(1 - p_j))$ (the exact marginal of the Dirichlet) and $\text{count}_{ij} \sim \text{Binomial}(n, \theta_{ij})$ (the exact marginal of the multinomial). This is done for the most abundant cell type and for a rare one (the rarest type with $c p_j \geq 0.25$; for the very rarest types a Monte Carlo SD is too noisy to compare with theory), with $N \in \{1, 10\}$ and n from 10^1 to 10^8 .

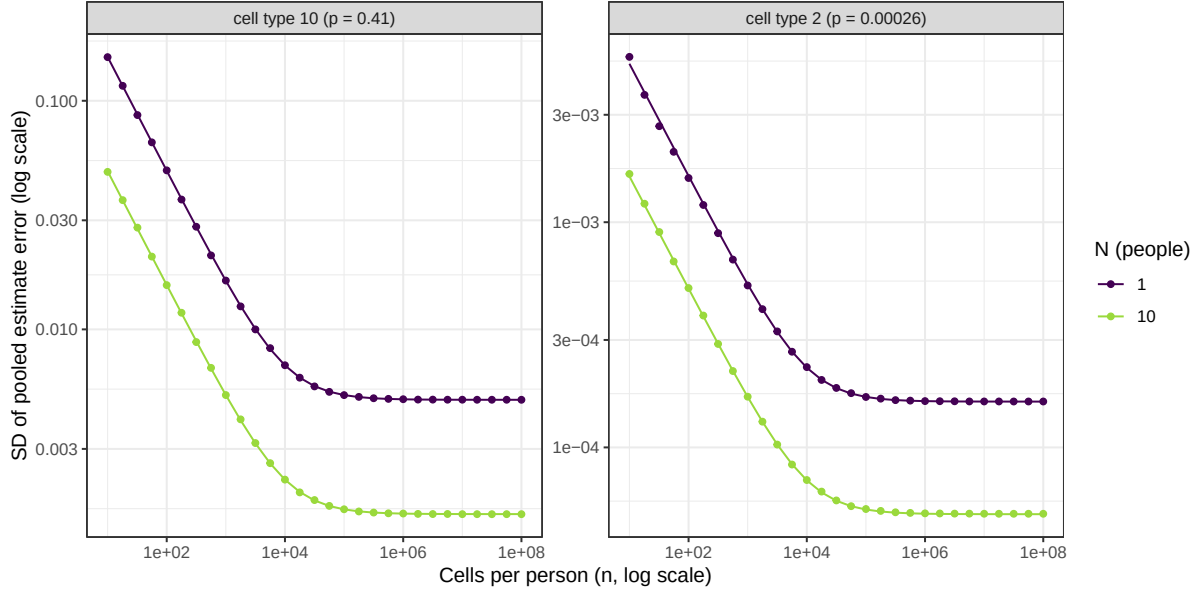


Figure 5: Standard deviation of $\bar{p}_j - p_j$ versus cells per person (both axes log scale) at $\alpha = 5$, $c = 10,000$. Points: simulated (20,000 replicates). Lines: Equation 3.

Figure 5 confirms the decomposition. The SD falls with n up to about $n \approx c$ and then flattens, and above roughly 10^5 cells adding cells no longer helps. The simulated SD stays within 1.2% of the formula for the most abundant type, and within 7.3% for the rare type ($c p_j \approx 2.6$).